

CAAM 31470 Lecture Notes (Spring 2025)

Motivation + Goals

Solutions to differential equations are difficult to obtain. While we can prove that they exist, formulas for them often do not exist or are so prohibitively complicated as to be useless.

In some situations this can be overcome by using complex analysis: the study of differentiable functions of a complex variable.

Example 1. Solving an ODE by Laplace Transform.

$$y'' + 4y' = \cos(t-3) + 4t, \quad y(0) = 2, \quad y'(0) = -1$$

While we know how to solve this equation by the method of undetermined coefficients, the process is tedious and requires trial and error. Instead, we can solve by Laplace transform.

Let $Y(s) = \int_0^\infty f(t)e^{-st} dt$. Using this

$$s^2 Y(s) - \underbrace{s y(0)}_{=2} - \underbrace{y'(0)}_{=-1} + 4(s Y(s) - \underbrace{y(0)}_{=2}) = \mathcal{L}(\cos(t-3) + 4t)$$

$$(s^2 + 4s) Y(s) - 2s + 1 - 8 = \frac{s^3 \cos(3) + s^2(4 + \sin(3)) + 4}{s^4 + s^2}$$

$$Y(s) = \left(2s + 7 + \frac{s^3 \cos(3) + s^2(4 + \sin(3)) + 4}{s^4 + s^2} \right) \frac{1}{s^2 + 4s}$$

That is, the differential equation becomes algebraic and we can solve for $Y(s)$.

For this method to be consistent, $\mathcal{L}$ needs to be an invertible map and we need to be able to find an inverse.

Example 2: Fourier Transforms

Computing Fourier transforms relies on being able to evaluate improper integrals of the form:

$$F(k) = \int_{-\infty}^{\infty} f(x) e^{ikx} dx$$

For example: $\int_{-\infty}^{\infty} \frac{e^{ikx}}{(x^2+1)^2} dx$

Example 3: Asymptotic Properties of Solutions

Airy's Equation: $\frac{d^2 y}{dx^2} - xy = 0$

has a solution $Ai(x) = \frac{1}{\pi} \int_0^{\infty} \cos\left(\frac{t^3}{3} + xt\right) dt$

No less complicated formula exists. However, we can determine properties of this function by complex analysis.

For very large x :

$$Ai(x) \approx \frac{1}{2\sqrt{\pi} x^{1/4}} \exp\left[-\frac{2}{3} x^{3/2}\right]$$

↑ this is not convergence, for more details see CAAM 31531 Asymptotic Analysis.

In each of these examples, the functions we wish to study are integrals of functions of real variables. However, just like studying polynomials is easier using complex numbers, it can be very helpful to study functions using complex number inputs.

Our goal in this class is to understand the basic properties of these functions and use them to our advantage in solving applied math problems.

Functions of a Complex Variable

Let $f: S \rightarrow \mathbb{C}$ where $S \subseteq \mathbb{C}$ is the domain of f .

Since both $z, f(z) \in \mathbb{C}$ we can also write

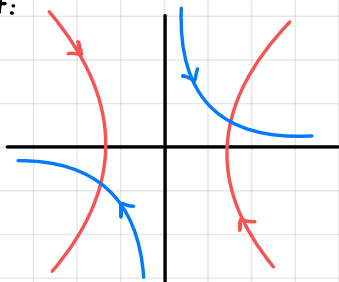
$$z = x + yi \quad \text{and} \quad f(z) = u + iv$$

where $u(x, y)$ and $v(x, y)$ are real-valued functions of $z = x + iy$.

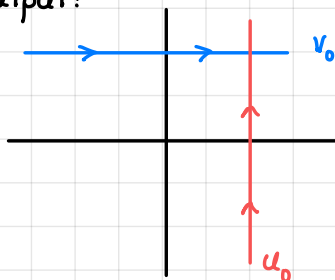
It is hard to visualize what these functions look like since both the inputs and outputs are two-dimensional. To get some idea we look at how curves are mapped to each other.

Example: $w = z^2$.

Input:



Output:

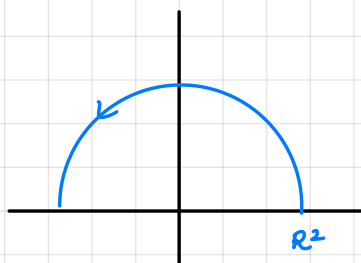
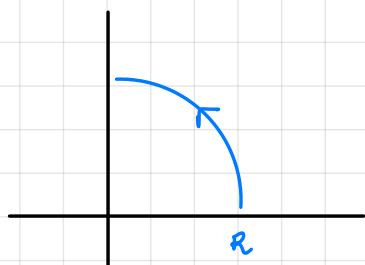


$$z^2 = (x + yi)^2$$

$$= x^2 - y^2 + 2xyi$$

$$\Rightarrow u = x^2 - y^2 \quad \text{and}$$

$$v = 2xy$$

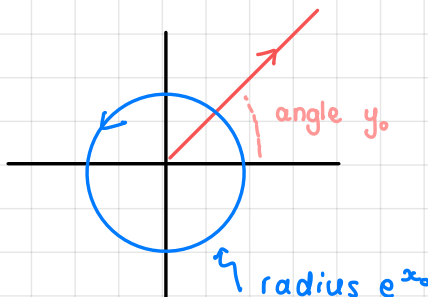
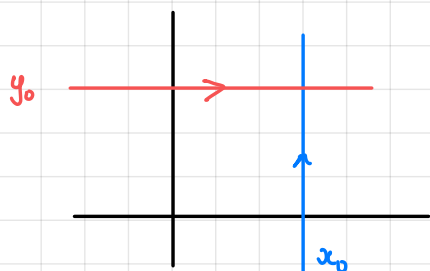


Alternatively

$$z^2 = (re^{i\theta})^2$$

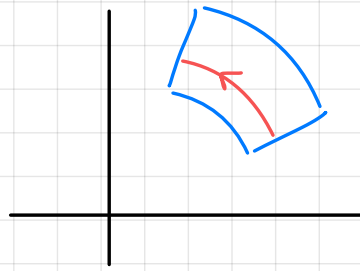
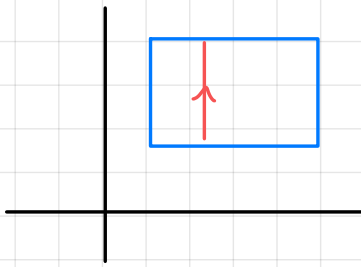
$$= r^2 e^{i2\theta}$$

Example: $w = e^z$



$$e^z = e^{x+iy}$$

$$= e^x e^{iy}$$



Limits and Continuity

Functions of a complex variable are naturally functions of two real variables. This means limits can be hard to achieve.

Example: Function on $\mathbb{R}^2$

Does the function: $f(x,y) = \frac{x^2 y}{x^4 + y^2}$ $(x,y) \neq (0,0)$.

have a limit as $(x,y) \rightarrow (0,0)$?

Answer is no! Note that for $x=0$ or $y=0$, $f(x,y)=0$.
Moreover, along any straight line

$$f(x, \alpha x) = \frac{\alpha x^3}{x^4 + \alpha^2 x^2} = \frac{\alpha x}{x^2 + \alpha^2}$$

which also has limit 0. However, along the lines $y = \pm x^2$:

$$f(x, \pm x^2) = \frac{\pm x^4}{x^4 + x^4} = \pm \frac{1}{2}.$$

Hence, f cannot have a limit as $(x,y) \rightarrow (0,0)$.

This is true in a similar way for functions of a complex variable.

Def: We say $\lim_{z \rightarrow z_0} f(z) = w_0$ if

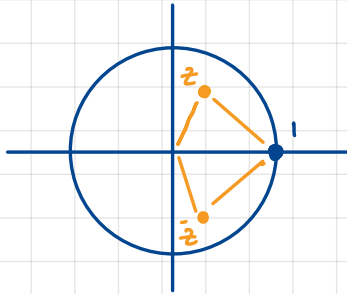
$$\forall \varepsilon > 0 \exists \delta > 0 \text{ s.t. } |f(z) - w_0| < \varepsilon \text{ whenever } |z - z_0| < \delta.$$

Moreover, f is continuous at z_0 if $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

Example: $f(z) = \frac{i}{2} \bar{z}$ for $|z| < 1$.

What is the limit as $z \rightarrow 1$ of $f(z)$?

Assuming $|z| < 1$: $|f(z) - \frac{i}{2}| = |\frac{i}{2} \bar{z} - \frac{i}{2}| = \frac{|\bar{z} - 1|}{2} = \frac{|z - 1|}{2}$



Hence, if $\delta = 2\varepsilon$ then

$$|z - 1| < \delta = 2\varepsilon \Rightarrow |f(z) - \frac{i}{2}| < \frac{2\varepsilon}{2} = \varepsilon.$$

$$\text{ie. } \lim_{z \rightarrow 1} f(z) = \frac{i}{2} = f(1).$$

Note: we only consider input values z inside the domain of definition.

Example: $f(z) = \frac{z}{\bar{z}}$, $z \neq 0$. Does $\lim_{z \rightarrow 0} f(z)$ exist?

Answer is no. It is enough to show different limits along two 1-dimensional paths.

$$\text{Let } z = x \Rightarrow f(x) = \frac{x}{x} = 1$$

$$z = yi \Rightarrow f(yi) = \frac{yi}{-yi} = -1$$

Hence, no limit can exist.

Thm: Suppose that $f(z) = u(x, y) + iv(x, y)$ for $z = x + yi$ and

$$z_0 = x_0 + iy_0 \quad \text{and} \quad w_0 = u_0 + iv_0.$$

Then:

$$\lim_{z \rightarrow z_0} f(z) = w_0 \quad (\Leftrightarrow) \quad \lim_{(x, y) \rightarrow (x_0, y_0)} u(x, y) = u_0$$

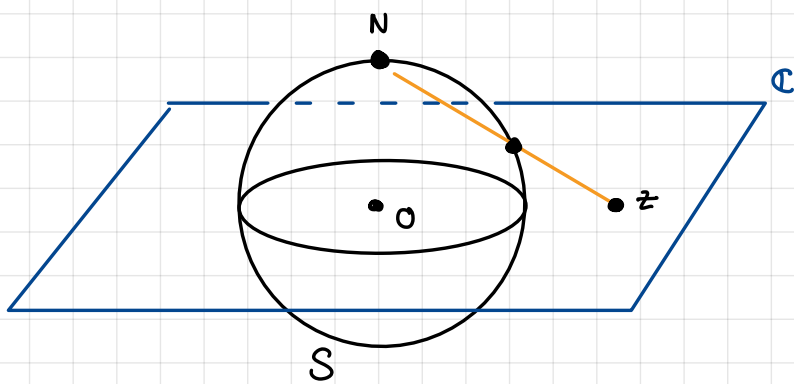
$$\text{and } \lim_{(x, y) \rightarrow (x_0, y_0)} v(x, y) = v_0.$$

The proof is left as a homework exercise.

Remark: From this we know using limits of functions on $\mathbb{R}^2$ that:

- limits are unique
- sum, product and quotient laws hold

We can also talk about limits at infinity by extending $\mathbb{C}$ to contain the "point" ∞ . One natural way to do this is using a sphere.



Let S be the unit sphere centred at $(0,0,0)$ in $\mathbb{R}^3$ and identify the xy -plane with $\mathbb{C}$.

Then each point $s \in S \setminus N$ can be mapped to a point $z \in \mathbb{C}$ by drawing a straight line from N through s and intersecting the plane.

The point N is then the point at ∞ in any direction.

Moreover, neighbourhoods of ∞ are the sets $\{z : |z| > \frac{1}{\varepsilon}\}$. Using this, we can use the regular definition of a limit.

Def: We say $\lim_{z \rightarrow \infty} f(z) = w_0$ if

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad |f(z) - w_0| < \varepsilon \quad \text{whenever} \quad |z| > \frac{1}{\delta}.$$

We say $\lim_{z \rightarrow z_0} f(z) = \infty$ if

$$\forall \varepsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} \quad |f(z)| > \frac{1}{\varepsilon} \quad \text{whenever} \quad |z - z_0| < \delta.$$

Thm: If $z_0, w_0 \in \mathbb{C}$ then

- $\lim_{z \rightarrow z_0} f(z) = \infty \iff \lim_{z \rightarrow z_0} \frac{1}{f(z)} = 0$.
- $\lim_{z \rightarrow \infty} f(z) = w_0 \iff \lim_{z \rightarrow 0} f\left(\frac{1}{z}\right) = w_0$.
- $\lim_{z \rightarrow \infty} f(z) = \infty \iff \lim_{z \rightarrow 0} \frac{1}{f\left(\frac{1}{z}\right)} = 0$.

Just like for functions of a real variable we also have the following properties of continuous functions.

Thm: • A composition of continuous functions is continuous.

• If a function $f(z)$ is continuous and non-zero at z_0 then $f(z) \neq 0$ throughout some neighbourhood of z_0 .

• If a function is continuous throughout a closed and bounded region R , then there exists a real value M s.t.

$$|f(z)| \leq M \quad \forall z \in R$$

and there exists at least one point $z_0 \in R$ s.t.
 $|f(z_0)| = M$.

Derivatives

So far it looks like functions of a complex variable are really very similar to functions on $\mathbb{R}^2$.

This changes when we start talking about differentiation.

Def: A function $f(z)$ is differentiable at a point $z_0 \in \mathbb{C}$ if the limit

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists.

Example 1. $f(z) = z^2$

$$\begin{aligned}\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} &= \lim_{z \rightarrow z_0} \frac{z^2 - z_0^2}{z - z_0} = \lim_{z \rightarrow z_0} \frac{(z + z_0)(z - z_0)}{z - z_0} \\ &= \lim_{z \rightarrow z_0} z + z_0 = 2z_0\end{aligned}$$

Hence, $f'(z) = 2z$.

Example 2: $f(z) = \bar{z}$ for $z_0 = 0$.

We want to evaluate: $\lim_{z \rightarrow z_0} \frac{\bar{z} - \bar{z}_0}{z - z_0}$ when $z_0 = 0$

But that is $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$ which we have seen does not exist.

ie. $f(z) = \bar{z}$ is not differentiable at $z = 0$.

In fact, f is not differentiable for any $z \in \mathbb{R}$.

Example 3: Let $f(z) = |z|^2 = x^2 + y^2$.

Recall that $|z|^2 = z\bar{z}$. So we have $f(z) = z\bar{z}$.

$$\begin{aligned}\text{Then } f'(z_0) &= \lim_{z \rightarrow z_0} \frac{z\bar{z} - z_0\bar{z}_0}{z - z_0} \\ &= \lim_{h \rightarrow 0} \frac{(z_0 + h)(\overline{z_0 + h}) - z_0\bar{z}_0}{h} \quad z = z_0 + h, h \in \mathbb{C} \\ &= \lim_{h \rightarrow 0} \frac{z_0\bar{z}_0 + z_0\bar{h} + h\bar{z}_0 + h\bar{h} - z_0\bar{z}_0}{h} \\ &= \lim_{h \rightarrow 0} z_0 \frac{\bar{h}}{h} + \bar{z}_0 + \bar{h}\end{aligned}$$

from which we see that the limit does not exist whenever $z_0 \neq 0$.

ie. $f'(0) = 0$ but is not defined for any other value.

In example 3, we have a function differentiable at a single point but nowhere else in a neighbourhood of that point.

On the otherhand when $f(z) = |z|^2$ we have

$$u(x,y) = x^2 + y^2 \quad \text{and} \quad v(x,y) = 0$$

both of which have infinitely many continuous partial derivatives with respect to x and y .

Differentiation by complex variables is substantially different to differentiation by real variables!

So what can we say about derivatives?

Thm: (Basic Derivatives)

Let $c \in \mathbb{C}$ be constant and f, g be differentiable functions of a complex variable z

Then the following properties hold.

$$(1) \quad \frac{d}{dz}(c) = 0, \quad \frac{d}{dz}(z) = 1, \quad \frac{d}{dz}(cf(z)) = cf'(z)$$

$$(2) \quad \frac{d}{dz}(z^n) = nz^{n-1}$$

$$(3) \quad \frac{d}{dz}(f(z) + g(z)) = f'(z) + g'(z)$$

$$(4) \quad \frac{d}{dz}(f(z)g(z)) = f'(z)g(z) + f(z)g'(z)$$

$$(5) \quad \frac{d}{dz}\left(\frac{f(z)}{g(z)}\right) = \frac{g(z)f'(z) - g'(z)f(z)}{(g(z))^2} \quad \text{when } g(z) \neq 0$$

$$(6) \quad \frac{d}{dz}(g(f(z))) = g'(f(z))f'(z)$$



note: g should be differentiable at $f(z_0)$
and f should be differentiable at z_0 .

That is, when we know functions are differentiable, we can use the normal rules of differentiation to evaluate derivatives.

So we still need criteria to determine when a function is differentiable.

Cauchy-Riemann Equations

Suppose $f(z) = u(x,y) + iv(x,y)$ is differentiable at a point $z = z_0$.

Then the following are true:

(1) u, v are continuous and have partial derivatives at (x_0, y_0) .

(2) $u_x = v_y$ and $u_y = -v_x$. ← Cauchy Riemann equations

By definition of the complex derivative:

$$f'(z_0) = \lim_{h \rightarrow 0} \frac{f(z_0+h) - f(z_0)}{h}, \quad h \in \mathbb{C}$$

where h is allowed to converge along any path.

Moreover, we have already seen that a complex limit exists if, and only if, the 2-dimensional real limits of the real and imaginary parts exist. That is, we know, for $h = x+iy$,

$$\lim_{(x,y) \rightarrow (0,0)} \frac{u(x_0+x, y_0+y) - u(x_0, y_0)}{x} - \left(\frac{v(x_0+x, y_0+y) - v(x_0, y_0)}{y} \right) \quad (\text{real part})$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{v(x_0+x, y_0+y) - v(x_0, y_0)}{x} + \left(\frac{u(x_0+x, y_0+y) - u(x_0, y_0)}{y} \right) \quad (\text{imaginary part})$$

exist. By taking paths along the real and imaginary axes, this shows that both u and v have partial derivatives.

In particular, taking $h = x$ ($y=0$):

$$\begin{aligned} f'(z_0) &= \lim_{x \rightarrow 0} \frac{f(z_0 + x) - f(z_0)}{x} \\ &= \lim_{x \rightarrow 0} \frac{u(x_0 + x, y_0) - u(x_0, y_0)}{x} + i \left(\frac{v(x_0 + x, y_0) - v(x_0, y_0)}{x} \right) \\ &= \frac{\partial u}{\partial x}(x_0, y_0) + i \frac{\partial v}{\partial x}(x_0, y_0) \end{aligned}$$

both individual limits
exist by the above
↓

Hence, we know both $\frac{\partial u}{\partial x}$ and $\frac{\partial v}{\partial x}$ exist at (x_0, y_0) .

Similarly, taking $h = iy$ ($x=0$):

$$\begin{aligned} f'(z_0) &= \lim_{iy \rightarrow 0} \frac{f(z_0 + iy) - f(z_0)}{iy} \\ &= \lim_{iy \rightarrow 0} \frac{u(x_0, y_0 + y) - u(x_0, y_0)}{iy} + i \left(\frac{v(x_0, y_0 + y) - v(x_0, y_0)}{iy} \right) \\ &= \frac{\partial v}{\partial y}(x_0, y_0) - i \frac{\partial u}{\partial y}(x_0, y_0) \end{aligned}$$

and so both $\frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y}$ exist as well.

Moreover, since both expressions equal $f'(z_0)$ we can equate them to find:

$$\begin{aligned} \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} &= \frac{\partial v}{\partial y} - i \frac{\partial u}{\partial y} \\ \Rightarrow \quad &\boxed{\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}} \end{aligned}$$

Whenever a function f is complex differentiable, its real and imaginary parts are related by the Cauchy-Riemann equations!

Example: $f(z) = z^2 = x^2 - y^2 + 2xyi$

$$u(x,y) = x^2 - y^2, \quad v(x,y) = 2xy$$

$$u_x = 2x \quad \begin{array}{c} \swarrow \text{green} \quad \searrow \text{blue} \end{array} \quad v_x = 2y$$

$$u_y = -2y \quad \begin{array}{c} \swarrow \text{blue} \quad \searrow \text{green} \end{array} \quad v_y = 2x$$

Example 2: $f(z) = |z|^2$ is not differentiable. We can use the Cauchy-Riemann equations to quickly check this.

$$u(x,y) = x^2 + y^2 \quad v(x,y) = 0$$

$$u_x = 2x \quad \begin{array}{c} \swarrow \text{green} \quad \searrow \text{blue} \end{array} \quad v_x = 0$$

$$u_y = 2y \quad \begin{array}{c} \swarrow \text{blue} \quad \searrow \text{green} \end{array} \quad v_y = 0$$

Hence, f can only be differentiable at $z=0$.

The Cauchy-Riemann equations are only a necessary condition for complex differentiability. To find a sufficient condition we also need some extra continuity.

Thm: Let $f(z) = u(x,y) + iv(x,y)$ be defined on a neighbourhood of $z_0 = x_0 + iy_0$ and suppose

- (a) the first-order partial derivatives of u, v with respect to x, y exist everywhere in the neighbourhood, and
- (b) the partial derivatives are continuous at (x_0, y_0) and satisfy the Cauchy-Riemann equations at (x_0, y_0) .

Then $f'(z_0)$ exists and its value is given by

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$